EcoLoop

AI-Assisted E-Waste Reuse & Recovery Marketplace

AI-powered digital platform for sustainable electronics diagnostics, recovery, and circular recycling.



Reuse



Repair



Recycle

TEAM MEMBERS

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01 · PLATFORM CONCEPT

About EcoLoop

PROBLEM

E-waste is hazardous, complex, and difficult to route responsibly.

Electronics contain valuable parts yet pose severe hazards (Lithium, Lead, Mercury). Users lack guidance on repairability, salvage, and safe recycling.

SOLUTION

One intelligent marketplace coordinates circular recovery.

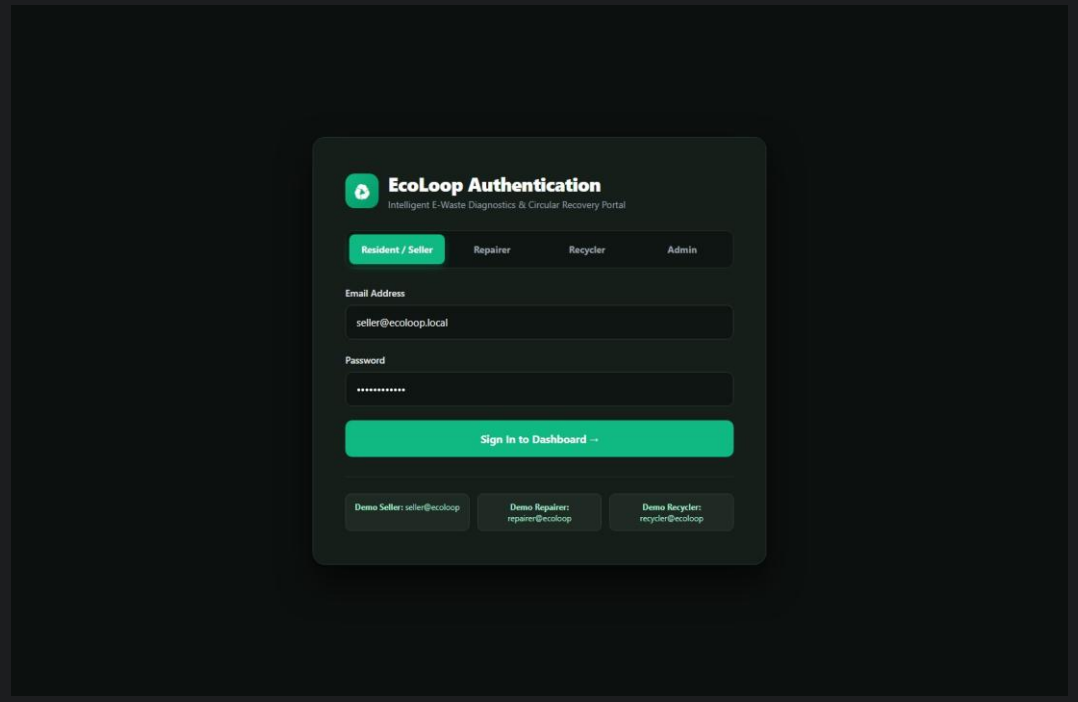
EcoLoop integrates MobileNetV2 vision for component detection, a deterministic decision engine, safety interlocks, and verified EcoImpact guidance.

HOW THE PROJECT WORKS

1. User uploads device photo and specifies basic condition.
2. MobileNetV2 AI detects device category and key internal/external components.
3. Interactive diagnostic questions evaluate power, screen, and battery safety.
4. Deterministic engine recommends optimal recovery route (Reuse, Repair, Refurbish, Parts, or Recycle).
5. Listed on marketplace connecting residents with verified repairers and certified recyclers.

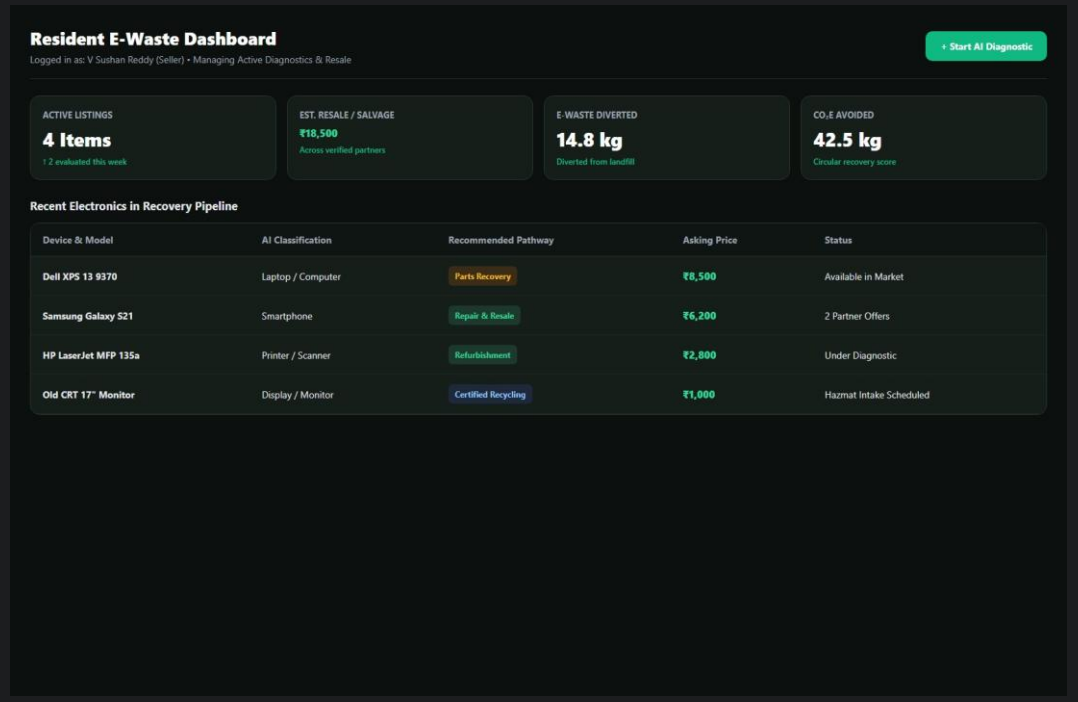
MODULE 1 · ACCESS AND USER OVERVIEW

Access and User Overview



User Authentication / Role-Based Access

Secure registration, login, and dedicated role portals for Residents/Sellers, Repairers, Recyclers, and Admins



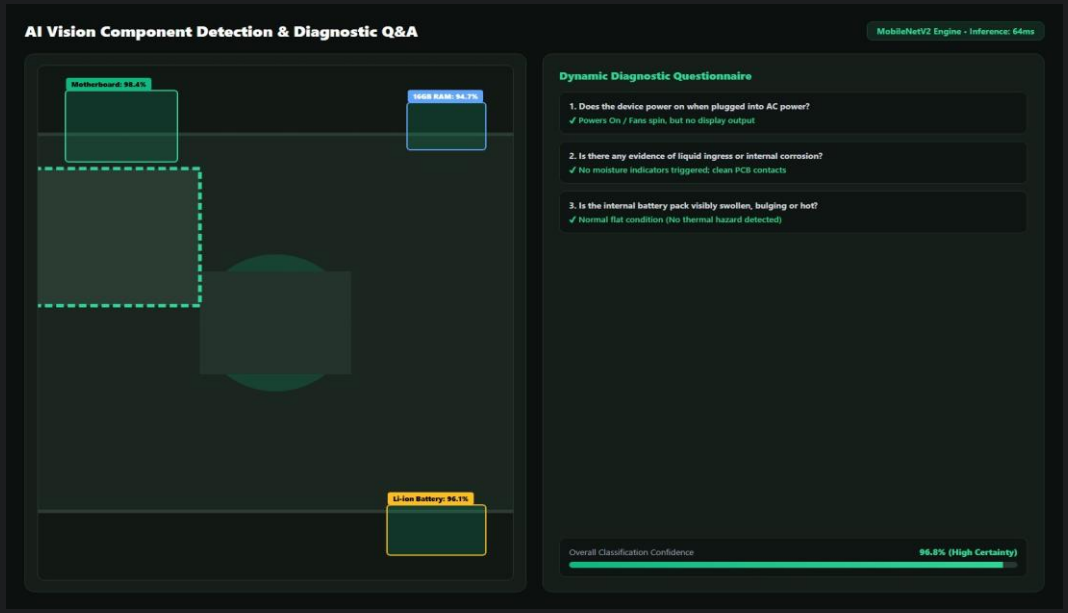
Member Dashboard / Overview

Active listings, residual valuations in ₹, recovery status tracking, and positive environmental impact footprint.

Entry and overview screens for managing a member's circular e-waste recovery journey. EcoLoop — E-Waste Marketplace.

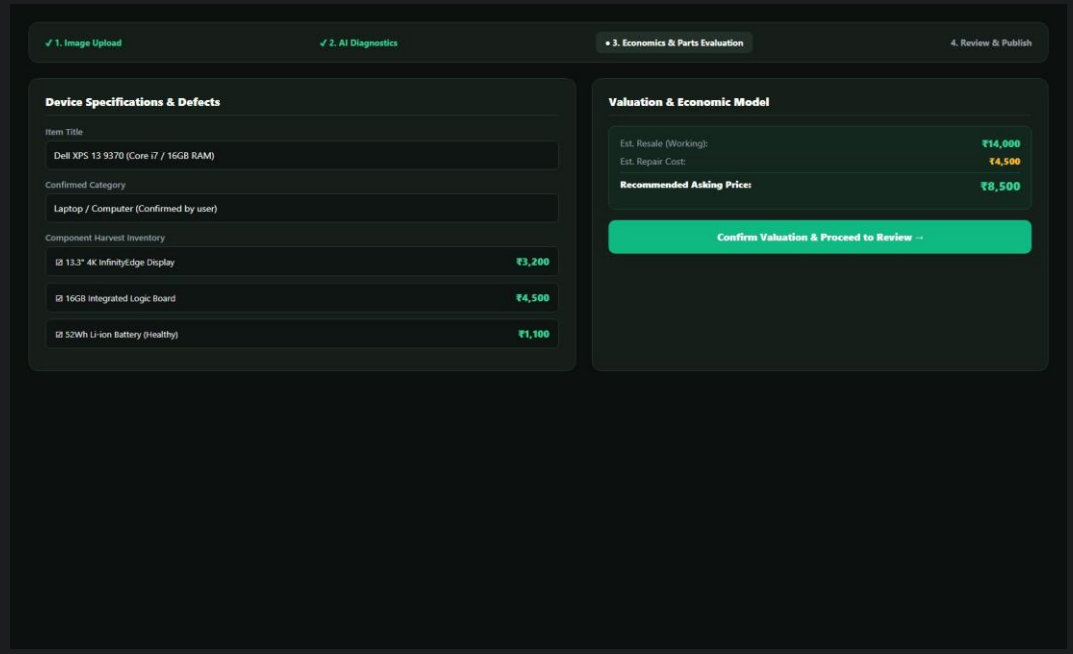
MODULE 2 · AI COMPONENT DETECTION & LISTING WIZARD

AI Component Detection & Listing Wizard



AI Vision & Component Diagnostics

MobileNetV2 deep learning classifier, automated component recognition, and interactive diagnostic Q&A.

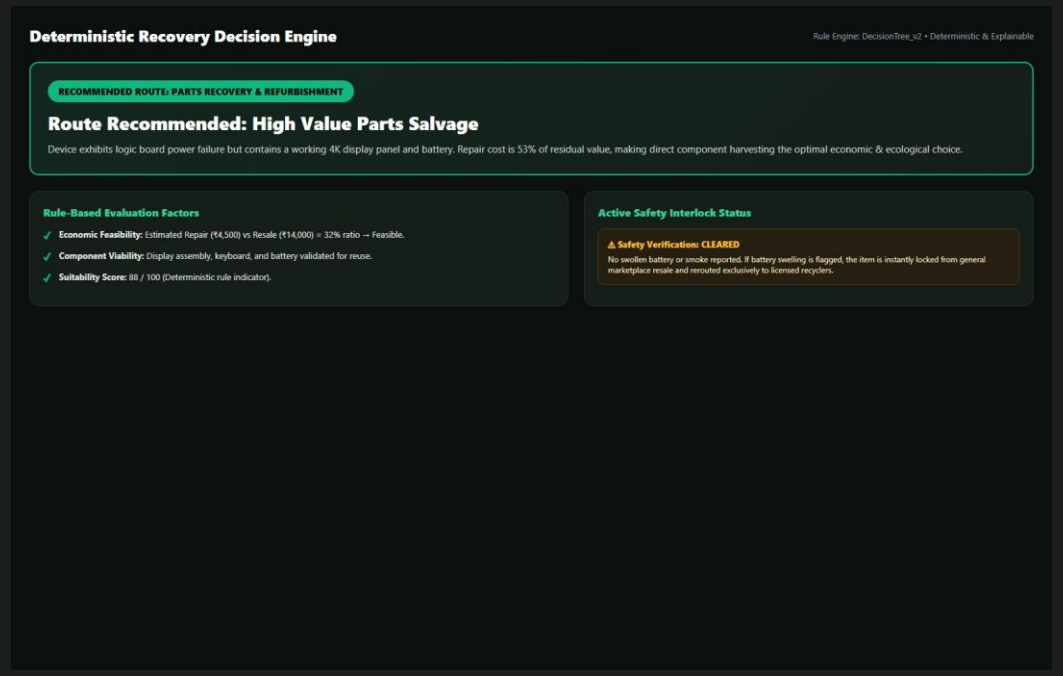


Multi-Step Listing Creation Wizard

Physical defect evaluation, parts breakdown, asking price in ₹, economic repair estimati

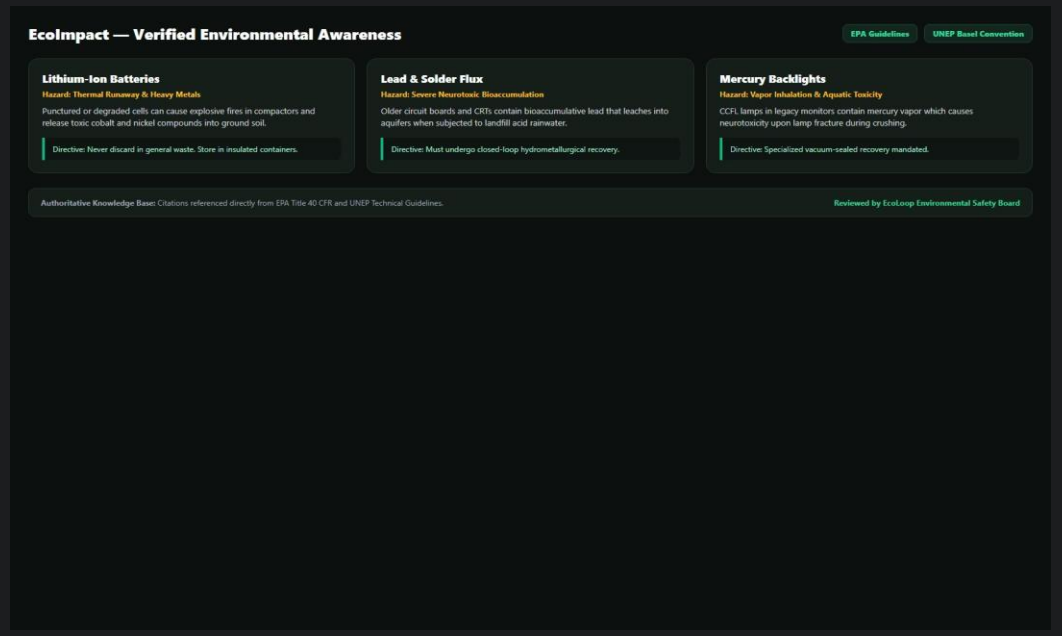
Members evaluate electronics with AI and publish verified listings. EcoLoop — E-Waste Marketplace.

Deterministic Recovery Engine & EcoImpact Hub



Recovery Decision Engine & Safety Interlock

Multi-tier economic feasibility rules, residual value evaluation, and immediate lockdown protocols for unsafe batteries.

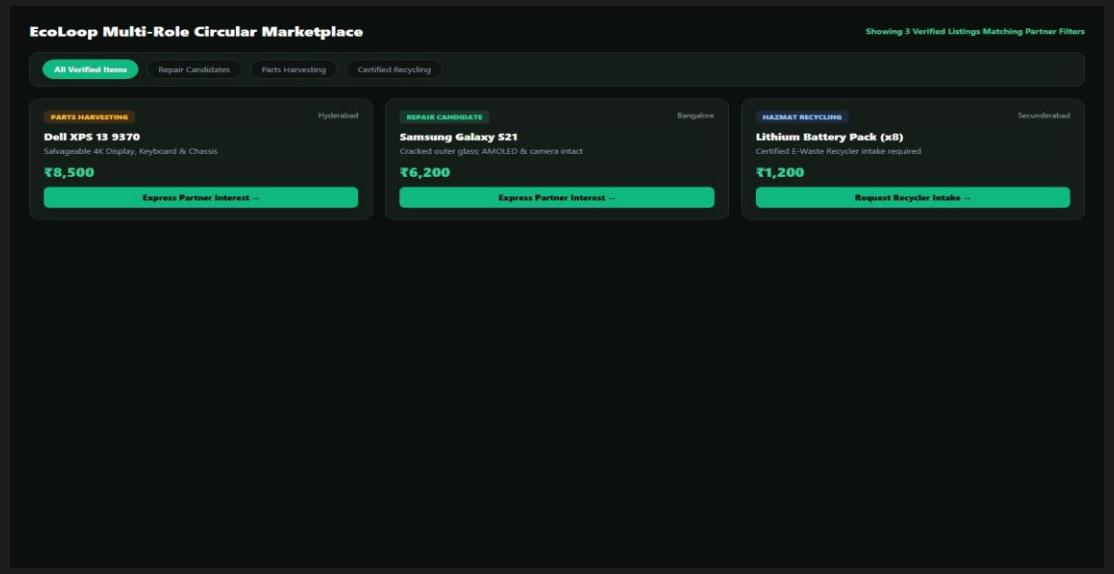


EcoImpact Environmental Knowledge Hub

Verified toxic substance hazards (Lithium, Lead, Mercury, BFRs), EPA/UNEP citations, and safe handling directives.

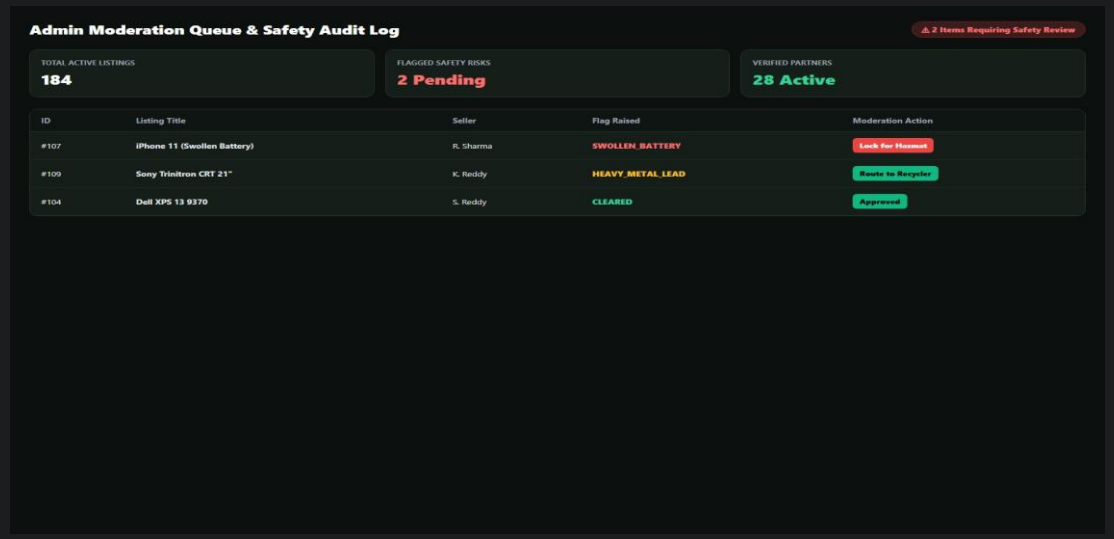
Deterministic recovery routing paired with verified environmental awareness. EcoLoop — E-Waste Marketplace.

Partner Operations and Administration



Repairer & Recycler Marketplace

Filter listings by route (Repair Candidates, Parts Harvesting, Certified Recycling), inspect component specs, and express partner interest.



Admin Dashboard & Moderation Queue

Moderation queue for flagged safety risks, status change audit trail, user governance, and process partner interest.

ECOLOOP

Thank You

Turning electronic waste into value, one recovery at a time.